

Modifications

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1 Semi-Automated Voxel-to-Point Transformation

1.1 Motivation

Traditional neuron reconstruction pipelines often convert voxel-based 3D images into point clouds using a fixed intensity threshold. This approach produces a dense, noisy set of points that typically fails to capture the neuron’s continuous structural skeleton—particularly the medial axis and critical branch junctions. Preserving these features is essential for accurate downstream processing (e.g., skeleton proposal and connectivity prediction). A semi-automated approach, which combines classical methods with automated refinement, offers both reliability and efficiency.

1.2 Proposed Approach

Our approach consists of two main components:

1. Skeleton Extraction (Manual/Predefined Step):

- Apply a classical skeletonization algorithm (e.g., TEASAR or Kimimaro) on the segmented 3D neuron volume.
- Extract the medial axis, which serves as a ground truth representation of the neuron’s core structure and branch junctions.
- Allow for manual verification or parameter adjustment to ensure the reliability of the extracted skeleton.

2. Automated Adaptive Sampling:

- For each candidate point (obtained from initial thresholding), compute the Euclidean distance to the nearest skeleton point.
- Transform these distances into sampling weights using an appropriate mapping (e.g., an exponential decay function), so that points closer to the medial axis receive higher weights.
- Employ a differentiable sampling mechanism (such as one based on Gumbel-softmax) to automatically select a refined subset of points.
- Integrate a geometric loss (e.g., Chamfer distance) that compares the refined point cloud to the extracted skeleton, thus supervising the adaptive sampling.

1.3 Implementation Details

Preprocessing:

- Start with the segmented 3D neuron image, obtained via intensity thresholding and normalization.
- Convert high-intensity voxels to an initial, dense point cloud.

Skeleton Extraction:

- Run the TEASAR or Kimimaro algorithm on the voxel volume to extract the medial axis.
- Optionally, allow for manual inspection and adjustment of parameters (e.g., threshold levels) to ensure the extracted skeleton accurately reflects the neuron’s structure.

Adaptive Sampling:

- For each point in the initial point cloud, compute its distance to the nearest skeleton point.
- Use these distances to calculate sampling weights (e.g., via an exponential decay function: $w = e^{-\alpha d}$, where d is the distance and α is a scaling parameter).
- Apply a differentiable sampling technique (e.g., Gumbel-softmax) to select a subset of points based on the computed weights.

Loss Integration:

- Introduce a geometric loss function, such as the Chamfer distance between the refined point cloud and the extracted skeleton.
- Optimize the parameters of the adaptive sampling module by minimizing this loss, ensuring that the output point cloud aligns well with the desired structural features.

1.4 Expected Benefits

This semi-automated approach offers several key advantages:

- **Reliability:** The classical skeleton extraction provides a dependable baseline that captures the neuron’s medial axis and branch junctions.
- **Automation with Oversight:** While the skeleton extraction step can be manually verified or tuned, the adaptive sampling module operates automatically and in a differentiable manner.
- **Structural Fidelity:** The refined point cloud is concentrated along the neuron’s true structure, reducing noise and redundant points.
- **Enhanced Downstream Performance:** A more informative point cloud leads to improved performance in subsequent modules such as feature encoding, skeleton proposal, and connectivity prediction.
- **Computational Efficiency:** By reducing non-informative points, the approach minimizes computational overhead in later stages.

1.5 Discussion

The semi-automated voxel-to-point transformation effectively bridges classical and modern techniques. The manual or predefined skeleton extraction ensures that the critical structural features are reliably identified, while the automated adaptive sampling refines the point cloud in a data-driven and differentiable manner. This balance allows for both expert oversight and efficient, scalable processing, setting a robust foundation for the overall neuron reconstruction pipeline.

2 Enhanced Point Cloud Encoder

2.1 Motivation and Overview

The original PointNeuron framework employs a DGCNN/EdgeConv encoder for extracting geometric features from the point cloud. However, recent advancements have produced architectures that achieve higher accuracy with improved efficiency. In this work, we propose replacing the original encoder with a state-of-the-art backbone—**PointNeXt**—and optionally integrating a lightweight transformer fusion module in later layers. This combination is designed to extract robust, multi-scale representations that capture both fine local geometry and long-range global context, thereby enhancing the performance of subsequent modules such as the skeleton proposal and connectivity predictor.

2.2 PointNeXt Backbone in Detail

- **Architecture Improvements:** PointNeXt revisits the classical PointNet++ design by introducing an inverted residual bottleneck structure and separable MLPs. This design minimizes parameter redundancy while preserving expressive power.
- **Local Feature Extraction:** The backbone employs farthest point sampling (FPS) and k-nearest neighbor (k-NN) grouping to define local neighborhoods. In each neighborhood, inverted residual bottleneck modules aggregate features, effectively capturing the fine details of neuronal branches as well as the overall morphology.
- **Advanced Training Strategies:** Extensive data augmentation, label smoothing, and state-of-the-art optimization techniques are utilized to enhance performance. These improvements not only boost accuracy but also reduce inference time compared to older methods like DGCNN.

2.3 Optional Transformer Fusion Module

- **Global Context Integration:** In deeper layers of the encoder, a lightweight transformer block can be integrated to capture long-range dependencies via self-attention. This module augments the local features extracted by PointNeXt with global contextual information.
- **Computational Efficiency:** The transformer is designed to be computationally lightweight, ensuring that it provides additional context without imposing significant overhead.

2.4 Integration and End-to-End Differentiability

The enhanced encoder takes as input the refined point cloud produced by the semi-automated voxel-to-point transformation module. The PointNeXt backbone (with the optional transformer module) outputs high-dimensional features for each point, which are then passed to the downstream skeleton proposal and connectivity prediction modules. The entire encoder remains end-to-end differentiable, ensuring that the network can be optimized jointly during training.

2.5 Expected Benefits

- **Enhanced Feature Quality:** Combining PointNeXt’s robust local feature extraction with the transformer’s global context results in multi-scale representations that accurately capture the complex geometry of neuron structures.

- **Improved Downstream Performance:** Richer features lead to more accurate detection of skeleton nodes and improved connectivity prediction, ultimately enhancing the overall neuron reconstruction.
- **Computational Efficiency and Scalability:** The efficient architecture and training strategies of PointNeXt offer significant speedup and reduced computational cost compared to older methods.
- **Flexibility:** The optional transformer module provides an additional performance boost in scenarios where global context is critical, while still maintaining a lean computational profile.

2.6 Discussion

Upgrading the encoder with PointNeXt, together with an optional transformer fusion module, constitutes a substantial improvement over previous approaches. The advanced design—characterized by inverted residual bottlenecks, separable MLPs, and robust training strategies—ensures effective capture of both local and global geometric features. This enhanced encoder is expected to drive superior performance in the subsequent stages of the neuron reconstruction pipeline, making it a vital component of the proposed PointNeuron2.0 system.

3 Advanced Skeleton Proposal Module

3.1 Motivation and Overview

The original PointNeuron framework employs an MLP-based regression combined with non-maximum suppression (NMS) to generate candidate skeleton nodes. This approach, however, may produce overlapping proposals or miss critical nodes in regions with sparse signals. To address these limitations, we reframe the skeleton proposal task as an object detection problem. In our proposed method, each skeleton node is treated as an object with associated properties (e.g., spatial coordinates, objectness confidence, and radius), enabling a more robust and precise detection process.

3.2 Proposed Approach

- **Detection Reformulation:** Instead of relying solely on regression, we treat the prediction of skeleton node coordinates as an object detection task. Each candidate is associated with properties such as spatial coordinates, an objectness score, and a radius.
- **Transformer Decoder Framework:** We employ a transformer decoder, inspired by the DETR framework, with a fixed set of learnable query embeddings. These queries attend to the high-dimensional features produced by the enhanced encoder and output candidate skeleton nodes. For each query, the decoder predicts:
 - The spatial coordinates (potentially with learned offsets),
 - An objectness score indicating whether the proposal represents a true skeleton node, and
 - A radius representing the local structural thickness.

This detection-style approach naturally handles overlapping proposals and removes the need for separate NMS post-processing.

- **Multi-Scale Detection:** To capture both coarse (trunk-level) and fine (branch-level) features, the transformer decoder integrates multi-scale queries. This design ensures reliable detection of nodes in both thick and thin regions.
- **Loss Functions and Matching:** The network is supervised with a composite loss comprising:
 - A classification loss (e.g., cross-entropy) for the objectness scores,
 - A regression loss (e.g., L2 loss) for the predicted coordinate offsets, and
 - A regression loss (e.g., Mean Absolute Error) for the predicted radii.

A set-based matching algorithm (e.g., the Hungarian algorithm) is employed during training to establish a one-to-one correspondence between the predicted proposals and the ground-truth skeleton nodes.

3.3 Implementation Details

- **Input:** The refined point cloud features, as obtained from the enhanced encoder, serve as input to the transformer decoder.
- **Transformer Decoder Architecture:** A fixed number of learnable query embeddings is initialized and passed through successive layers of multi-head self-attention and cross-attention. This enables the queries to aggregate both local and global contextual information, resulting in a set of candidate skeleton node proposals.
- **Loss Supervision:** The decoder is optimized by minimizing a composite loss that combines the classification, coordinate regression, and radius regression terms. This loss ensures that the decoder produces high-quality, non-redundant proposals that closely match the ground-truth skeleton.

3.4 Expected Outcomes and Benefits

- **Higher Precision in Skeleton Detection:** The transformer-based detection framework yields more accurate and distinct skeleton proposals, reducing overlap and enhancing localization accuracy.
- **Robustness Across Scales:** The integration of multi-scale queries enables the network to capture both large-scale (trunk) and fine-scale (branch) features, ensuring robust detection across varied neuron structures.
- **Streamlined Pipeline:** By reframing skeleton proposal as a detection problem, the need for post-hoc clustering or NMS is eliminated, simplifying the overall pipeline.
- **Enhanced Downstream Performance:** More reliable and precise skeleton proposals provide a strong foundation for the connectivity prediction module, ultimately improving the overall neuron reconstruction accuracy.

3.5 Discussion

The proposed advanced skeleton proposal module transforms the original regression-based approach into a detection framework, leveraging a transformer decoder to generate high-quality candidate nodes. By incorporating multi-scale detection and a composite loss that enforces both classification and regression accuracy, this module is expected to significantly improve skeleton localization. In

turn, the improved proposals will facilitate better connectivity prediction, thereby enhancing the fidelity of the final neuron reconstruction.

4 Improved Connectivity Prediction

4.1 Motivation and Overview

Accurate neuron reconstruction requires not only the detection of key skeleton nodes but also the correct prediction of their connectivity. In the original framework, a graph autoencoder was used to predict connectivity; however, this approach may fail to capture subtle geometric cues and enforce a proper tree structure. Our approach explicitly constructs a graph from the detected skeleton nodes and employs a graph neural network module to predict edge probabilities. By integrating both learned features and geometric information—and by incorporating topology-aware losses—we aim to produce connectivity predictions that conform to a biologically plausible tree structure.

4.2 Proposed Approach

- **Graph Construction:**

- Construct a graph where each node corresponds to a detected skeleton point.
- Establish initial candidate edges based on spatial proximity (e.g., via k-nearest neighbors or a distance threshold).

- **GATv2-based Connectivity Module:**

- Employ a Graph Attention Network v2 (GATv2) to predict the likelihood of connections between nodes.
- Utilize the dynamic, query-conditioned attention mechanism of GATv2 to weigh candidate edges by considering both the differences in learned features and explicit geometric cues (such as Euclidean distance, relative orientation, and radius differences).

- **Incorporation of Geometric and Topological Features:**

- For each potential edge, combine local feature differences with geometric relations to compute a robust edge score.

- **Topology-Aware Losses:**

- Apply a standard cross-entropy loss over the predicted edge probabilities against the ground-truth connectivity (extracted from SWC files).
- Introduce regularization terms that penalize cyclic connections, multiple parent assignments, or disconnected nodes to enforce a tree-like topology.

4.3 Implementation Details

- **Input Features:** The connectivity module receives refined node features from the advanced skeleton proposal module. These features are augmented with each node’s spatial coordinates and predicted radius.

- **GATv2 Architecture:** A stack of GATv2 layers is used, where each layer updates node representations by computing attention weights over neighboring nodes based on both feature differences and geometric relationships.

- **Loss Integration:** The module is optimized with a composite loss function:
 - A cross-entropy loss on the predicted edge probabilities.
 - Topology-aware regularizers that penalize deviations from the expected tree structure.

4.4 Expected Outcomes and Benefits

- **Accurate Connectivity Prediction:** The GATv2-based module dynamically weighs candidate edges, yielding precise and robust edge predictions.
- **Biologically Plausible Graphs:** The incorporation of topology-aware losses ensures that the predicted graph adheres to the expected tree structure of neurons.
- **Enhanced Reconstruction Fidelity:** Reliable connectivity predictions improve the final SWC output, leading to an overall higher-fidelity neuron reconstruction.

4.5 Discussion

By explicitly modeling edge relationships using a GATv2 architecture and integrating both learned features and geometric cues, the proposed connectivity prediction module addresses the limitations of prior autoencoder-based methods. The use of topology-aware loss functions further enforces the tree structure, resulting in more accurate and biologically meaningful neuron reconstructions.

5 End-to-End Integration and Training Strategy

5.1 Overview and Motivation

Our goal is to integrate the modified modules into a unified, end-to-end trainable pipeline. This pipeline comprises:

1. The semi-automated voxel-to-point transformation module.
2. The enhanced point cloud encoder (using the PointNeXt backbone with an optional lightweight transformer fusion module).
3. The advanced skeleton proposal module based on a DETR-inspired transformer decoder.
4. The improved connectivity prediction module employing a GATv2-based graph module with topology-aware losses.

Joint training of these modules allows for simultaneous optimization, ensuring that the final neuron reconstruction is both geometrically accurate and topologically consistent.

5.2 Pipeline Integration

- The refined point cloud from the semi-automated voxel-to-point transformation serves as the input.
- The enhanced encoder extracts robust, multi-scale features from each point.
- The skeleton proposal module uses these features to generate candidate skeleton nodes.

- The detected nodes are then organized into a graph, which the connectivity prediction module processes to infer the correct connections.
- Finally, the predicted node parameters and connectivity are interpreted to generate an SWC file representing the reconstructed neuron.

5.3 Loss Functions and Optimization

The network is trained with a composite loss that includes:

- **Sampling Loss:** A geometric loss (e.g., Chamfer distance) to ensure the refined point cloud aligns with the extracted ground-truth skeleton.
- **Detection Loss:** A combination of:
 - A cross-entropy loss for the objectness scores,
 - An L2 regression loss for coordinate offsets, and
 - A regression loss (e.g., Mean Absolute Error) for the predicted radii.
- **Connectivity Loss:** A cross-entropy loss on the predicted edge probabilities, supplemented with topology-aware regularization terms to penalize cycles, multiple parent assignments, or disconnected components.

The overall loss is a weighted sum of these components, ensuring balanced optimization across the pipeline.

5.4 Training Strategy

- **End-to-End Training:** The entire network is trained jointly via backpropagation.
- **Data Augmentation:** Employ robust augmentation techniques such as random rotations, scaling, and jittering to improve generalization.
- **Optimization:** Use modern optimizers such as AdamW with cosine learning rate decay and label smoothing to enhance convergence.
- **Hyperparameter Tuning:** Key parameters (learning rate, weight decay, loss weights, etc.) are tuned on a validation set to achieve optimal performance.

5.5 Expected Outcomes and Benefits

- **Joint Optimization:** End-to-end training enables the model to learn optimal representations and inter-module interactions, leading to improved overall reconstruction fidelity.
- **Improved Reconstruction:** The composite loss drives each module to perform optimally, yielding an SWC output that closely reflects the true neuron structure.
- **Efficiency and Scalability:** By leveraging efficient components (PointNeXt, lightweight transformers, GATv2) alongside modern training strategies, the pipeline reduces computational overhead and scales effectively to large datasets.

5.6 Discussion

Integrating all components into a single trainable framework allows the model to exploit synergistic effects between geometric and topological losses. This unified approach removes the need for separate post-processing steps (such as clustering or NMS), thereby simplifying the architecture while enhancing both speed and accuracy in neuron reconstruction.

6 Evaluation and Future Work

6.1 Evaluation Metrics

We plan to evaluate the overall neuron reconstruction using a combination of quantitative metrics that assess both geometric fidelity and topological correctness. These metrics include:

- **F1-score:** To evaluate the precision and recall of detected skeleton nodes.
- **Path Distance Score (PDS):** To measure the accuracy of reconstructed paths relative to the ground truth.
- **Euclidean Score (ESA):** To assess the spatial accuracy of the predicted skeleton.
- **Chamfer Distance:** To measure the similarity between the predicted and ground-truth skeleton point sets.

6.2 Datasets

The primary evaluation will be conducted on the Janelia-Fly dataset from the BigNeuron project, which features diverse neuron morphologies. This dataset will be split into training, validation, and testing subsets. Additionally, we plan to explore the pipeline’s performance on other neuron datasets or related 3D reconstruction tasks to validate its generalizability.

6.3 Experimental Setup

- **Training Protocols:** End-to-end training will be performed using cross-validation to ensure robustness.
- **Hyperparameter Tuning:** Key hyperparameters such as the learning rate, weight decay, and loss weight factors will be tuned on a validation set.
- **Hardware and Runtime:** Experiments will be conducted on high-performance computing resources to ensure scalability.
- **Ablation Studies:** Detailed ablation studies will be carried out to isolate the contributions of each module (adaptive sampling, enhanced encoder, advanced skeleton proposal, and connectivity prediction) and to assess the impact of the optional transformer fusion module.

6.4 Results

- **Quantitative Analysis:** We expect improvements in reconstruction metrics (higher F1-scores, lower PDS/ESA errors) compared to baseline methods.
- **Qualitative Analysis:** Visualizations of the reconstructed neuron skeletons will demonstrate improved continuity and connectivity.

- **Ablation Results:** Experimental results will detail how each modification contributes to overall performance and help identify optimal configurations.

6.5 Future Work

- **Enhanced Transformer Integration:** Explore deeper or multi-stage transformer fusion to further enhance global context understanding.
- **Semi-/Self-Supervised Training:** Investigate methods to reduce reliance on annotated data by leveraging semi-supervised or self-supervised learning techniques.
- **Multi-Modal Data Fusion:** Integrate additional modalities (e.g., image intensities, radius gradients) to further enrich feature representations.
- **Advanced Topological Methods:** Explore extended persistence or multifiltration techniques to capture more complex topological features, potentially improving connectivity predictions.
- **Scalability and Applications:** Adapt the pipeline for larger datasets and extend the approach to related domains such as cardiac surface reconstruction or other 3D medical image analyses.

6.6 Discussion

This evaluation framework is designed to rigorously test the proposed pipeline. By employing a diverse set of metrics and performing thorough ablation studies, we aim to validate that the improvements in voxel-to-point transformation, feature encoding, skeleton proposal, and connectivity prediction collectively yield a higher-fidelity neuron reconstruction. The insights gained from this evaluation will guide future enhancements, addressing any limitations and further improving model performance.

7 Conclusion

In this work, we have presented **PointNeuron2.0**—a comprehensive, end-to-end pipeline for 3D neuron reconstruction that integrates several key modifications to enhance both geometric and topological accuracy. Our semi-automated voxel-to-point transformation module leverages classical skeleton extraction in combination with a learnable adaptive sampling process, ensuring that the resulting point cloud faithfully represents the neuron’s medial axis and critical branch junctions.

Building on this foundation, our enhanced point cloud encoder replaces the original DGCNN/EdgeConv backbone with the state-of-the-art **PointNeXt** architecture, augmented with an optional lightweight transformer module. This design enables robust multi-scale feature extraction, capturing both fine local geometric details and long-range contextual information.

Furthermore, we reformulated the skeleton proposal stage by adopting a DETR-inspired transformer decoder. This approach recasts the detection of skeleton nodes as an object detection task, yielding high-quality, non-redundant candidate proposals that are essential for accurate neuron reconstruction. Finally, our improved connectivity prediction module utilizes a GATv2-based graph neural network to predict edge probabilities, integrating both learned features and explicit geometric cues, and employing topology-aware loss functions to enforce a biologically plausible tree structure.

Overall, **PointNeuron2.0** represents a significant advancement in automated 3D neuron reconstruction, providing a robust, efficient, and scalable solution for large-scale neuron analysis.